

💡 **HARIKA'S STRATEGY:** You deliver the entire opening arc – from the real disaster hook, through the 4 failure breakdowns, to the 4 engineering solutions and performance numbers. You establish complete authority, then execute the power handoff to Nikhil for technical deep dives and Q&A defense.

SLIDE 01

Opening Hook & Project SUTRA Introduction

00:00 - 00:30 (30s)

[Stand center stage • Take a 1-second breath • Deliver opening hook with serious, grounded tone]

🔥 **HOOK:** "When over 300 lives were buried under mud in the **July 2024 Wayanad landslides**, and flash floods cut off entire mountain valleys across **Kedarnath and Nepal**, rescue teams had only 72 hours to save lives. But the moment standard drones entered thick forests and deep ravines, they lost signal and crashed."

"Respected jury, we are **Team OFFGRID**. I am Harika, and with me is Nikhil, our Tech Lead. We built **Project SUTRA** — an autonomous drone swarm engineered to find survivors even when **GPS fails and normal communications collapse**."

PILLAR 1

Self-Flying Swarms: Drones dodge trees and coordinate together at 50 times a second without ground towers. [50Hz GNC]

PILLAR 2

Smart AI Video: Live video stays connected through deep mountain valleys without blacking out. [-5dB JSCC]

PILLAR 3

Thermal & 3D AI Vision: Spots survivors through smoke and finds their exact coordinates on steep slopes. [3D DEM • 3.59cm]

SLIDE 02

The Problem: 4 Real Disaster Breakdowns

00:30 - 01:05 (35s)

[Click to Slide 2 • Switch tone to urgent, factual disaster audit]

⚠️ **THE AUDIT:** "When our team audited actual NDRF and Army rescue reports from recent disaster zones, we found 4 major breakdowns:"

BREAKDOWN 1

Trees Block GPS: In Wayanad, thick tree cover blocked satellite signals, causing **70% of drones to crash into branches**.

BREAKDOWN 2

Mountains Block Radio: In mountain gorges, steep ridges blocked radio signals, dropping video feeds to total black.

BREAKDOWN 3

Slope Errors: On steep 45-degree hillsides, standard drone cameras assumed flat ground, **missing survivor locations by 15 to 30 meters**.

BREAKDOWN 4

Pilot Overload: Deploying 5 separate drones required **15 to 25 crew members** and 2 hours of setup, costing ₹12.5 Lakhs per flight.

SLIDE 03

The Solution: 4 Hardened Moats & Performance Delta

01:05 - 01:50 (45s)

[Click to Slide 3 • Deliver solutions with confident clarity • Point to performance table]

🛡️ **THE SOLUTION:** "SUTRA solves each breakdown through 4 hardened engineering moats:"

MOAT A

Self-Flying Swarms: Drones calculate collision-free flight paths onboard at 50 times a second — zero ground towers, zero GPS reliance.

MOAT B

Smart AI Video Link: When signals drop behind mountain ridges, our neural link keeps the video running down to -5dB without blacking out.

MOAT C

Thermal & 3D AI Vision: Combines body heat sensors with 3D elevation raycasting on Jetson Orin (<15ms), **completely removing the 30-meter slope error**.

MOAT D

1-Person Command Tablet: An offline 3D tactical map allows **just 1 operator to fly the entire swarm**, cutting mission costs to **₹95,000**.

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi ²	2-3 hours	45-60 min	10-18 min (4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000

"The bottom line: SUTRA searches a full square mile in just 10 minutes — 4 times faster than human rescue crews, saving lives when every second counts."

👉 POWER HANDOFF TO NIKHIL:

"Now, our Tech Lead, Nikhil, will walk you through the deep engineering behind each subsystem and our real-world deployment."